

Moaaz Emam Ahmed

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EDUCATION

Cairo University || Faculty of Engineering || GPA: 3.58

2022-2027

Major: Communication and Computer Engineering

SKILLS

- **Programming Languages:** C++, C, C#, Python, JavaScript, SQL, ARM assembly, Verilog HDL, Java
- **Development Tools & Environments:** Git, GitHub, Linux, Visual studio, IntelliJ IDEA, SQL Server, MongoDB, NodeJS, Postman, .NET framework, ModelSim, Logisim, Apache Airflow, Docker, Scrapy, Playwright, n8n, Airtable
- **Relevant Coursework:** Algorithms & Data Structures, Computer Networks, Machine Intelligence, Compilers, Operating Systems, Databases
- **Soft Skills:** Problem solving, Team collaboration, Rapid adaptability and learning, Effective communication
- **Certifications & Courses:**
 - Backend Development Certificate – IEEE
 - AI Certificate – IEEE

EXPERIENCE AND PROJECTS

Data Engineering Intern: *Instacodigo (Remote)*

Jul 2025 – Oct 2025

- Designed and implemented an **automated data pipeline** using **Apache Airflow** for large-scale web data collection, processing, validation, and integration with **Odoo ERP**.
- Built a modular scraping framework using **Scrapy** and **Playwright** to extract structured data from **static** and **JavaScript-rendered** websites at scale.
- Implemented **data cleaning**, deduplication, serialization, and **schema enforcement using Pydantic**, with branching **DAG workflows** routing products, jobs, and articles to appropriate Odoo models via OdooRPC.
- Developed an automated social media workflow using **n8n** and **Airtable**, supporting scheduled content verification and **posting to 13+ platforms** with **OAuth2 authentication** and token refreshing.

Google Docs Clone: *Advanced programming techniques*

2025

- Developed a real-time collaborative text editor in **Java** using **Spring Boot**, **WebSockets**, and **JavaFX**, supporting **multi-user editing** with role-based access.
- Implemented concurrent edits using a custom **CRDT** algorithm, multithreading, and asynchronous updates.
- Integrated features like import/export files, user tracking with colored cursors using **JavaFX**, user comments, and graceful disconnection and reconnection.

Hospital Information Website: *Backend development course*

2025

- Developed and implemented a scalable hospital management system using **Node.js**, **Express** and **MongoDB Atlas**, allowing ease of access for patients, doctors and admins.
- Developed a secure authentication system using **JWT** and role-based access control for different user types, ensure privacy and secure access to sensitive information.
- Conducted unit and integration testing using **Jest** and **Supertest** to validate API functionality and system performance.

Custom Programming Language Compiler: *Compilers*

2025

- Designed and implemented a full compiler front-end for a custom programming language using **Flex** and **Bison** in C.
- Built lexical analysis, parsing, AST construction, and semantic analysis supporting variables, functions, control flow, loops, scoping, and basic data types.
- Designed **grammar rules**, symbol table, and **three-address code (TAC) intermediate representation**.
- Implemented **semantic checks** including type mismatch, undeclared/redeclared identifiers, and function parameter mismatch, and developed a **GUI-based code editor** for compilation and visualization.

OS Scheduler and Memory Manager: *Operating systems*

2024

- Developed an OS-level scheduler in C on a **Linux** environment that selects among 4 implemented scheduling algorithms: SJF, Preemptive HPF, RR (with configurable quantum), Preemptive MLFQ.
- Engineered a process generator to simulate task arrivals and integrated intercross communication using message queues, signals, and shared memory, while employing forking to handle concurrent processes.
- Designed and implemented a Buddy Memory Allocation system using a tree structure with recursive splitting and merging for efficient dynamic memory management and minimized fragmentation.

Fashion House Management System: *Databases*

2024

- Designed and implemented a multi-user windows application using **C#** and **.NET** framework, implementing event-driven programming and interactive GUI.
- Engineered a secure **SQL** relational database in SQL Server Management studio, designing ER diagrams and optimizing table relationships for efficient data retrieval and improved application responsiveness.
- Developed secure authentication system with hashed passwords and encrypted payment details to ensure safe card storage and transaction processing.

Alien vs Earth War simulation: *Data structures and Algorithms*

2024

- Developed a simulation for a conflict between two armies using **C++**. Optimized for large-scale simulations.
- Designed and implemented **custom data structures** for various army elements to efficiently manage different real-time operations such as insertion, deletion and updates.
- Engineered multiple simulation modes driven by advanced algorithms to compute battle outcomes and generate comprehensive statistics on the battle results.

Automated Garbage Truck with Material Segregation: *Microprocessors*

2025

- Designed, built, and programmed a fully autonomous garbage-picking car using **STM32 Blue pill MCU** and **ARM Assembly**.
- Implemented **obstacle avoidance** using servo-mounted **sensors** to scan surroundings, garbage detection and metal-nonmetal segregation using metal detectors, and bin-level detection with an embedded return-to-drop-site mechanism.
- Led the team in **hardware implementation and integration**, **CAD** arm design, and optimized sizing and weight distribution, ensuring optimal performance.

Digital ALU: *Logic Design*

2024

- Designed and implemented both software and hardware version of a custom 3 signed-bit ALU supporting 4 modes: addition, subtraction, multiplication, and remainder division.
- Developed the ALU using **Verilog HDL** and rigorously tested it using **ModelSim** and testbenches.
- Designed hardware prototype using **Logisim** and transitioned the design to physically optimized implementation to reduce component count and resource usage to achieve maximum space and cost efficiency.